

plete without mentioning ZigBee and Bluetooth, and it is no wonder that these have their own groups promoting them, ZigBee Alliance and Bluetooth SIG, respectively.

ZigBee Alliance works to connect devices of dramatically different natures into a single network, and in any environment, with control in the hands of the user. An open, non-profit association, promoters of this group include NXP, Philips, Schneider Electric, Texas Instruments and Silicon Labs, while the list of members includes ARM, Atmel, Analog Devices, Blackberry, Cisco Systems, Microchip and Samsung.

ZigBee Alliance offers three specifications for enabling interoperability, namely, ZigBee PRO specification as a foundation that concentrates on low power and network interconnection, ZigBee RF4CE specification for bi-directional communication applications and ZigBee IP specification for supporting IP based standards. With ZigBee gateways, it is easier for devices to adopt this standard.

With Bluetooth Smart finding wide acceptance, especially as a bridging technology in existing IoT devices, Bluetooth SIG is also focusing on developing the standards behind Bluetooth. The group, which boasts of members like Ericsson, Intel, Lenovo, Microsoft, Apple and Toshiba, aims to make it the preferred low-power wireless networking technology for the IoT. Having seen the success of mesh networks in Bluetooth, they are working towards formally defining a global industry standard for Bluetooth Smart mesh networks.

Woolley says, "Smart mesh networks will provide for a very long-range capability, with devices spread across potentially huge areas in large buildings or even urban areas. It will introduce the possibility of having redundant paths between devices, making sure there is a backup in case a device

fails, like in all mesh networks. But what makes Bluetooth Smart mesh networks all the more compelling is that these are based on a technology that virtually all smartphones have today. Mobile computing in many forms will be an important aspect of the IoT and so building IoT networks using Bluetooth immediately opens up some exciting possibilities, given the simple fact that it is so widespread."

A hoard of operating systems

With the IoT being all about connecting devices together, how do we

For devices working on other OSes, a compatibility library is to be installed for interoperability. Together, Brillo and Weave aim to provide a complete platform for connected devices.

The two Lite systems. Lite OS, supported by National Science Foundation grant, is an open source, interactive real-time OS designed for wireless sensor networks and embedded systems. With a UNIX-like environment, programming is hybrid, allowing both event-driven and thread-driven codes.

ARM mbed

The ARM mbed IoT device platform provides the operating system, cloud services, tools and developer ecosystem to make the creation and deployment of commercial, standards based IoT solutions possible at scale. Applications for the platform can be developed using mbed online integrated development environment, editing the code and compiling it via the cloud. The core software platform also includes ports that will allow mbed to run on microcontrollers by different manufacturers.

mbed boards that have been designed for IoT purposes almost invariably have some form of connectivity mechanism either built-in or by means of an extension board. ARM mbed transport layer security (TLS) forms the core of mbed communication security strategy and ARM aims at a combination of multi-layer hardware protection and a software ecosystem to secure connected devices.

Mikko Saarnivala, VP - technology, IoT division, ARM, tells us what the transport layer security in ARM mbed OS means for users and developers. He says, "From the user point of view, this does not really show up in any way. The only thing the user needs to know is that mbed OS-enabled devices have state-of-the-art transport layer security and crypto mechanisms in place. For developers who are working on new IoT devices, services and products, this means that there is a standard Internet [TLS/datagram transport layer security (DTLS)] security mechanism that we can use to secure traffic all the way from the constrained IoT devices to the cloud services and servers."

leave out the OSes these work on?

Google' Weave through Brillo devices. Brillo, the Android based embedded OS platform by Google, is essentially a lower version of Android and will integrate easily with Android devices. It supports Bluetooth Low Energy (BLE), Wi-Fi and Thread, and works with IoT devices that are low-power and memory-constrained.

Along with Brillo, Google has also introduced Weave, a protocol to aid intercommunication between Brillo and other IoT devices. It will act as a common framework for applications to use these devices.

Huawei has also come up with an ecosystem for connected objects, Agile IoT, which includes a lightweight system called LiteOS, a network gateway and a dedicated controller to link the chain. This needs just 10kB, which is less than that occupied by a normal document and, hence, is correctly described as an ultra-light OS.

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